

Passive RF-Silent Drone Detection via Neuromorphic Vision and Acoustic Sensor Fusion of the Propeller Blade-Pass Frequency

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Abstract—We investigate passive, RF-silent drone detection by exploiting the fact that a spinning propeller emits a characteristic blade-pass frequency (BPF) simultaneously in two physical channels: optical flicker, sensed by a neuromorphic event camera, and acoustic pressure, sensed by a microphone array. We present a working prototype that (i) extracts per-pixel flicker frequency from a Prophesee EVK4-HD event camera and isolates propeller regions via connected-component spatial clustering and temporal tracking, (ii) extracts the acoustic BPF and a direction-of-arrival (DOA) estimate from a four-microphone array using multi-channel FFT and GCC-PHAT, and (iii) fuses the two streams with a sequential Bayesian updater that includes harmonic-aware frequency cross-validation. The central finding is methodological: detection *architecture*, not parameter tuning, determines success. A grid-FFT baseline failed at range regardless of tuning, whereas replacing it with per-pixel frequency extraction and connected-component clustering produced stable, low-false-positive detection, with first-detection latency reduced from 600–1200 ms to approximately 100 ms after real-time optimization. Acoustic BPF detection and cross-modal frequency agreement worked reliably; cross-modal *spatial* (azimuth) calibration did not converge to a usable model under our indoor test conditions ($R^2 = 0.137$), which we report as an instructive negative result.

Index Terms—event camera, neuromorphic vision, drone detection, counter-UAS, sensor fusion, microphone array, blade-pass frequency, Bayesian inference

I. INTRODUCTION

Consumer drones are inexpensive, increasingly autonomous, and—critically for defense applications—can fly with no active radio-frequency (RF) link by navigating GPS waypoints. This defeats RF scanning, the dominant commercial drone-detection technology. Each of the remaining passive and active modalities has a structural blind spot, summarized in Table I: radar struggles to separate small drones from birds, frame-based cameras suffer motion blur and limited dynamic range, and acoustic-only systems exhibit short range and false-positive rates of 10–20 % in ambient noise.

A. Research Question and Hypothesis

We ask: can a *passive* sensor combination exploit the physics-guaranteed propeller blade-pass frequency (BPF) to detect drones with a false-alarm rate far below acoustic-only systems, without any RF emission?

Our hypothesis is that because the BPF appears simultaneously in the optical and acoustic domains, requiring *cross-modal frequency agreement* should suppress the dominant

TABLE I
STRUCTURAL WEAKNESSES OF EXISTING DETECTION MODALITIES

Modality	Mechanism	Structural weakness
RF scanning	Detect drone radio protocol	Blind to autonomous / RF-silent drones
Radar	Reflected RF	Cannot separate small drones from birds; high cost
RGB / thermal	Frame imaging (30–60 fps)	Motion blur, ~60 dB dynamic range, fails in low light / fog
Acoustic only	Propeller sound	Short range; 10–20 % false-positive rate in noise

false-alarm sources—each of which usually manifests in only one modality—driving the joint false-positive probability sharply down.

B. Contributions

This paper makes the following contributions:

- 1) A working prototype, to our knowledge the first, that fuses a neuromorphic event camera with a microphone array for drone detection via the shared BPF observable.
- 2) An architectural ablation demonstrating that per-pixel frequency extraction with connected-component spatial clustering succeeds where a grid-FFT baseline fails at range regardless of parameter tuning—a transferable lesson for event-based detection pipelines.
- 3) A harmonic-aware cross-modal frequency validation scheme and a gated sequential Bayesian fusion framework, with observed accumulate/decay dynamics matching design intent.
- 4) Real-time engineering techniques (downscaled morphology, pre-allocated buffers, region-of-interest statistics) that reduce first-detection latency by 6–12×.
- 5) An honest account of negative results, including the failure of visual-to-acoustic azimuth calibration to converge indoors, and the unsuitability of a speech-oriented DOA framework (ODAS) for impulsive propeller acoustics.

II. RELATED WORK

Event-based vision. Neuromorphic event cameras asynchronously report per-pixel log-intensity changes with microsecond latency and >120 dB dynamic range [1], [2]. Mod-

ern sensors such as the Sony IMX636 provide HD resolution in a stacked back-illuminated process [3]. Their temporal resolution makes them natural instruments for measuring high-frequency periodic phenomena such as vibration and rotating machinery—the property we exploit here.

Counter-UAS detection. Surveys of drone detection cover RF, radar, electro-optical, and acoustic approaches [4], [5]. Acoustic systems detect the propeller’s harmonic signature but suffer short range and high false-alarm rates in noise; radar and RF systems are active or blind to RF-silent targets. We are not aware of prior work fusing an event camera with a microphone array for counter-UAS purposes.

Acoustic DOA. Generalized cross-correlation with phase transform (GCC-PHAT) [6] remains the standard lightweight time-difference-of-arrival estimator; ODAS [7] provides an integrated localization/tracking stack tuned primarily for speech.

Light-flicker positioning. A companion line of work uses luminaire flicker as a location fingerprint [8]–[10]; we discuss a preliminary feasibility study in Section X.

III. PHYSICAL PRINCIPLE

A propeller with B blades rotating at shaft rate f_s (Hz) modulates both reflected light and radiated sound at the blade-pass frequency

$$f_{\text{BPF}} = B \cdot f_s = \frac{B \cdot \text{RPM}}{60}. \quad (1)$$

Optically, each blade sweep changes local scene luminance, generating events in a neuromorphic sensor at f_{BPF} (and harmonics). Acoustically, the pressure field carries the same fundamental and its harmonics. Because the two channels share a single physical cause, requiring their measured frequencies to agree (up to harmonic ambiguity, Section VI) rejects noise sources that are periodic in only one modality—flickering luminaires, wind noise, speech, or vibrating structures.

IV. VISUAL DETECTION PIPELINE

The visual pipeline went through nine documented iterations; the crucial step was an architectural replacement, not tuning.

A. Baseline: Grid-Based FFT (v1) — Failure at Range

The first detector divided the 1280×720 sensor into a 16×16 grid, counted events per cell per $500\mu\text{s}$ slice, ran a rolling-window FFT per cell, thresholded on SNR, and clustered adjacent active cells. This worked only at centimeter range: at distance the propeller occupies a handful of pixels inside a 160×90 cell crowded with noise pixels, so its spectral peak is diluted below the cell noise floor.

B. Parameter Tuning — A Key Negative Result

To chase range, thresholds were relaxed aggressively (filter length $7 \rightarrow 4$, maximum period difference $500 \rightarrow 1500\mu\text{s}$, minimum pixel count $25 \rightarrow 5$, grid $16 \rightarrow 8$ cells, FFT window $0.5 \rightarrow 1.0$ s, minimum SNR $3.0 \rightarrow 2.0$). The result was 25–43 “propeller regions” per frame scattered across random frequencies—pure noise promoted to false positives.

The grid FFT has no notion of spatial coherence: it never asks whether the active pixels in a cell are spatially or spectrally consistent. *Parameter tuning cannot repair a fundamentally wrong architecture.*

C. Per-Pixel Frequency + Connected Components (v2)

The redesigned pipeline is

per-pixel freq. map $\rightarrow$ binary mask $\rightarrow$ dilation
 $\rightarrow$ connected components
 $\rightarrow$ freq.-consistency filter $\rightarrow$ tracking.

Three design choices proved decisive:

- **Spatial clustering instead of a fixed grid.** Connected-component labeling adapts to the propeller’s true shape; even 2–3 connected pixels at a consistent frequency are meaningful, while the same pixels scattered across a grid cell are not.
- **Frequency-consistency gate.** Within each cluster the coefficient of variation of per-pixel frequency must satisfy $\text{CV} \leq 0.3$.
- **Temporal tracking.** Greedy association on spatial proximity (< 100 px) and frequency similarity ($< 30\%$); confirmation only after a minimum number of consecutive hits; pruning after 8 idle frames; exponential smoothing ($\alpha = 0.3$) of position and frequency.

On a 3-blade bench propeller this yielded a single stable track at ~ 99.7 Hz ($\approx 2,990$ blade passes per minute equivalent), locked position, ~ 100 active pixels, and 270+ consecutive hits over the recording, with transient single-pixel false positives pruned within 1–2 s.

D. Ablation: Minimum Cluster Size

Setting the minimum cluster size to a single pixel produced 13–17 “confirmed propellers” per frame at random frequencies, confirming that *spatial coherence (≥ 2 connected pixels) is the primary false-positive filter*: a single pixel carries no spatial information and is indistinguishable from sensor noise. The final configuration (minimum 2 pixels, 5 confirmation hits) yielded clean detections with zero false positives and multi-pixel clusters (35–269 px) tracked over 80+ hits in a 32 s recording.

E. Real-Time Optimization (v3)

The synchronous frequency-map callback initially ran full-resolution morphology, connected-component analysis, and a frame copy at 25 Hz, costing ~ 315 ms of CPU per wall-clock second and causing monotonically growing display lag. Four fixes eliminated the drift: (1) $4\times$ downscaling ($1280 \times 720 \rightarrow 320 \times 180$) before morphology; (2) reducing the update rate to 10 Hz; (3) in-place frame drawing; and (4) early exit on near-empty masks, reducing callback cost to ~ 20 ms/s. A subsequent production pass added pre-allocated buffers (eliminating garbage-collection jitter), region-of-interest-only frequency statistics, velocity-predictive tracking, and a Bayesian

TABLE II
VISUAL DETECTOR PERFORMANCE, v2 vs. v3

Metric	v2	v3	Gain
First detect (strong signal)	600–1200 ms	~100 ms	6–12×
First detect (weak signal)	600–1200 ms	~200 ms	3–6×
Steady-state update	100–140 ms	50–90 ms	~2×
Per-frame analysis	~1.5 ms	~0.8 ms	~2×
GC jitter	±5 ms	< ±1 ms	~5×

confidence score replacing the hard hit-count threshold: each observation contributes likelihood

$$p = 0.65 + q(0.85 - 0.65), \quad (2)$$

where $q \in [0, 1]$ combines pixel-count and frequency-consistency sub-scores, accumulated as

$$P = 1 - (1 - p)^n \quad (3)$$

over n observations and decayed by $0.8\times$ per idle frame. Table II summarizes the measured effect.

V. ACOUSTIC DETECTION PIPELINE

A. BPF Extraction

A rolling 0.2 s window (3200 samples at 16 kHz) per channel is Hann-windowed and transformed; magnitude spectra are incoherently averaged across the four channels (~6 dB SNR gain). The peak is picked in the target band, its SNR computed against the median noise floor, and detections below a minimum SNR of 3.0 rejected. This proved reliable on sustained propeller tones; source calibration on the bench propeller measured a dominant tone at 162.7 Hz with SNR ≈ 24 dB.

B. DOA via GCC-PHAT

For each opposing microphone pair (64 mm baseline) we compute the phase-transform cross-spectrum

$$R_{12}(\tau) = \int \frac{X_1(\omega)X_2^*(\omega)}{|X_1(\omega)X_2^*(\omega)|} W(\omega) e^{j\omega\tau} d\omega, \quad (4)$$

with a frequency weighting $W(\omega)$ emphasizing the target band, $4\times$ zero-padded inverse transform for sub-sample delay resolution, parabolic peak interpolation, and a peak-to-second-peak confidence-ratio gate (≥ 1.5). Pair estimates are fused by averaging when both are confident, followed by a 7-sample median filter; the sample spread gates unstable bearings. This yields usable azimuth on steady tones; the planar 32 mm-radius array cannot resolve elevation and provides $\sim \pm 10^\circ$ azimuth accuracy at range.

C. ODAS — A Negative Result

We integrated the ODAS framework [7] via a custom socket relay. It produced robust DOA on speech (sustained utterances of 500 ms or more) but was unreliable on propeller acoustics, which are comparatively impulsive and broadband. This motivated the custom GCC-PHAT implementation above. The array’s on-board DSP additionally exposes a firmware DOA with speech-tuned voice-activity detection; it was likewise unreliable on steady tones and was not fused.

VI. SENSOR FUSION

A. Harmonic-Aware Frequency Cross-Validation

One modality frequently reports the blade-pass tone while the other reports the shaft fundamental or a harmonic. We therefore search harmonic pairs (n_v, n_a) , $1 \leq n_v, n_a \leq 4$, and declare a match if

$$\frac{|f_v/n_v - f_a/n_a|}{f_a/n_a} < 0.05, \quad (5)$$

returning confidence $1/(n_v n_a)$ so that fundamental-to-fundamental agreement outweighs high-order coincidences.

B. Sequential Bayesian Updating

Starting from prior $P(D) = 0.001$, each modality contributes a likelihood-ratio update with sigmoid likelihoods:

$$P(V | D) = \sigma(\text{SNR}_v - 5), \quad P(V | \neg D) = 0.01 e^{-0.5 \text{SNR}_v}, \quad (6)$$

$$P(A | D) = \sigma(\text{SNR}_a - 4), \quad P(A | \neg D) = 0.05, \quad (7)$$

with the acoustic false-alarm likelihood reduced to 0.005 when the noise floor is elevated. A cross-modal frequency match contributes $P(F | D) = 0.99 \cdot \text{conf}$ against $P(F | \neg D) = 0.001$. When no evidence arrives, the posterior decays toward the prior by $0.995\times$ per frame. A drone is declared at $P > 0.8$.

C. Fusion Gating

To prevent overconfident fusion from coincidental evidence, several gates proved necessary in practice: a visual quality gate (minimum pixel count and track confidence ≥ 0.98); an acoustic stability gate (persistence and frequency jitter ≤ 25 Hz); a temporal association gate ($|t_v - t_a| \leq \sim 200$ ms); an angle-residual gate (visual azimuth vs. acoustic DOA within 35°); and stricter audio-only gates (SNR ≥ 6 dB and bounded DOA spread) before an audio-only update may raise $P(D)$. In a live run with no propeller in view, the detector repeatedly latched $P(D) \rightarrow \sim 1.0$ on strong 140–200 Hz tones through the audio-only path and decayed back when the source stopped—the accumulate/decay dynamics behaved as designed, and the run illustrates that single-modality dominance is a real failure mode that the gates must police.

VII. IMPLEMENTATION

Visual. Prophesee EVK4-HD (Sony IMX636, 1280×720 , $\sim 1 \mu\text{s}$ per-pixel temporal resolution, >120 dB dynamic range) [3], with a 220° fisheye lens for hemispheric coverage and a rectilinear C-mount lens (63.8° HFOV) for azimuth experiments; Metavision SDK frequency map algorithms. **Acoustic.** ReSpeaker 4-mic circular USB array (32 mm radius, 64 mm opposing-pair baseline), 16 kHz / 16-bit capture; on-board XMOS XVF-3000 DSP and a 12-LED ring used as a physical bearing indicator. Pixel-to-azimuth mapping uses a pinhole model, $f_{\text{px}} = (W/2)/\tan(\text{HFOV}/2)$, $az = \text{atan2}(x - W/2, f_{\text{px}})$. **Software.** Python 3.9 (required by the SDK binaries), NumPy, OpenCV, PyAudio. The visual Nyquist

TABLE III
CONSOLIDATED OUTCOMES

Method / component	Outcome
Per-pixel flicker-frequency extraction	✓
Connected-component spatial clustering	✓
Temporal tracking + Bayesian confidence	✓
Real-time latency engineering	✓
Acoustic BPF via multi-channel FFT	✓
GCC-PHAT azimuth on sustained tones	✓
Harmonic-aware frequency cross-validation	✓
Gated sequential Bayesian fusion	✓
Grid-FFT architecture (v1)	×
Parameter tuning to extend range	×
Single-pixel minimum cluster size	×
ODAS for propeller acoustics	×
Visual→acoustic azimuth calibration	×
Elevation DOA (planar array)	×

constraint is $f_{\max} \leq 10^6 / (2 \Delta t)$ for accumulation period Δt in μs (1 kHz at the default $\Delta t = 500 \mu\text{s}$). Session traces are logged as JSONL with per-interval posterior, evidence, association lag, and gate states.

VIII. EXPERIMENTS AND RESULTS

All experiments used a bench propeller as a controlled rotating source and a synthetic on-screen vibration generator (15–120 Hz) for end-to-end validation of the visual frequency pipeline. Recordings were indoor, single-source, and $\sim 30\text{ s}$ long.

A. Summary of Outcomes

Table III consolidates what worked and what failed across the study.

B. Azimuth Calibration — A Negative Result

To validate the angle-residual gate we attempted to fit a linear map $\text{DOA}_a = g \cdot \text{az}_v + b$ by sweeping a co-located source laterally, binning samples by visual azimuth with robust per-bin medians, and fitting with iterative MAD outlier rejection and an R^2 solidity check (threshold 0.6). The best fit (rectilinear lens, 63.8° HFOV) achieved $g = -0.011$, $b = 0.37^\circ$, $R^2 = 0.137$, RMSE 0.70° —not solid. The acoustic DOA spanned only $\sim \pm 1.2^\circ$ while the visual sweep spanned $\sim \pm 25^\circ$, consistent with indoor reverberation, an insufficiently wide or slow sweep, or low GCC-PHAT confidence during the session. Cross-modal *spatial* fusion therefore remains unvalidated; cross-modal *frequency* fusion is validated.

IX. DISCUSSION AND LESSONS LEARNED

Architecture beats tuning. The single most consequential change in the project was architectural: replacing grid-FFT event counting with per-pixel frequency extraction and connected-component clustering. No amount of threshold relaxation rescued the grid detector; it merely promoted the noise floor to false positives.

Spatial coherence is the false-positive filter. The single-pixel ablation demonstrates that requiring as few as two connected pixels at a consistent frequency removes essentially

TABLE IV
LIMITATIONS AND THREATS TO VALIDITY

Area	Limitation
Dataset	Indoor, single source, $\sim 30\text{ s}$ recordings
Range	No outdoor / GPS ground-truth tests
Spatial fusion	Azimuth calibration weak ($R^2 = 0.137$)
DOA	Planar 32 mm array; $\sim \pm 10^\circ$, no elevation
Multi-target	One source per cluster assumed
Synchronization	No hardware clock; $\sim 100\text{ ms}$ uncertainty
Harmonics	Blade-count / shaft-rate ambiguity
FPR claim	Design-derived, not yet measured

all sensor-noise detections. Persistence (temporal tracking) removes roughly 95 % of the remaining transients.

Modality dominance is a real failure mode. The audio-only run in Section VI shows that a Bayesian fuser will happily saturate on one strong modality; explicit quality, association, and angle gates—not the update rule itself—are what make fusion trustworthy.

Tool-domain mismatch. ODAS and the array’s firmware DOA, both tuned for speech, failed on propeller acoustics—a reminder that off-the-shelf audition stacks encode source assumptions that must be checked before integration.

X. LIMITATIONS AND FUTURE WORK

Table IV lists the principal threats to validity. All detection results are from indoor, short, single-source lab recordings; no outdoor, multi-drone, or GPS-ground-truth validation was performed, so range figures are projections rather than measurements, and the sub-0.1 % false-positive-rate target is design-derived and must be established empirically on a labeled dataset. The 32 mm planar array cannot resolve elevation, harmonic ambiguity (a 2-blade propeller at 100 Hz BPF is indistinguishable from a 4-blade at 50 Hz shaft rate) requires external metadata, and there is no hardware inter-sensor clock ($\sim 100\text{ ms}$ association uncertainty).

Future work includes outdoor validation with GPS ground truth, a non-planar or larger-baseline array for elevation, hardware time synchronization, multi-target tracking, and repeating the azimuth calibration outdoors with a wider, slower sweep. A companion feasibility study on using ceiling-light flicker frequencies as location fingerprints—inspired by LiTell [8], iLAMP [9], and event-camera visible-light positioning [10]—completed a literature review and utility implementation with a positive feasibility verdict, but end-to-end validation remains open.

XI. CONCLUSION

We demonstrated a passive, RF-silent drone-detection prototype that fuses neuromorphic vision and acoustic sensing through the shared blade-pass frequency observable. The validated contributions are the architectural comparison (per-pixel frequency plus connected components decisively outperforms grid FFT), the real-time engineering that brings first-detection latency to $\sim 100\text{ ms}$, harmonic-aware cross-modal frequency

validation, and gated sequential Bayesian fusion whose accumulate/decay dynamics behave as designed. We report the failure of indoor visual-to-acoustic azimuth calibration as a genuine, instructive negative result rather than an omission. Range and false-positive-rate figures remain design targets pending outdoor validation on labeled data.

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